

Report of Analysis using Python

Great! Here's a clean and structured **Exploratory Data Analysis (EDA) Report** for the **Titanic dataset** using Python. You can use this for documentation, presentation, or as a base for a notebook/report submission.

Titanic Dataset - Exploratory Data Analysis Report

1. Objective

To explore the Titanic dataset to uncover patterns, relationships, and insights that could help understand factors influencing passenger survival.

2. Dataset Overview

- **Dataset:** train.csv from Kaggle's Titanic dataset
 - **Shape:** (891, 12)
 - **Target Variable:** Survived (0 = No, 1 = Yes)
-

3. Data Dictionary (Key Columns)

Column	Description
PassengerId	Unique ID
Survived	Survival (0 = No, 1 = Yes)
Pclass	Ticket class (1 = 1st, etc.)
Name	Name of passenger
Sex	Gender
Age	Age in years
SibSp	# of siblings/spouses aboard
Parch	# of parents/children aboard
Ticket	Ticket number

Column	Description
Fare	Passenger fare
Cabin	Cabin number (many missing)
Embarked	Port of Embarkation (C/Q/S)

□ 4. Data Cleaning

- **Missing Values:**
 - Age: 177 missing → filled with median
 - Embarked: 2 missing → filled with mode
 - Cabin: 687 missing → dropped
 - **Duplicates:** None found
-

5. Univariate Analysis

Survival Count

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```
sns.countplot(x='Survived', data=df)
```

- More passengers **did not survive** (549) than survived (342)

Gender Distribution

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```
sns.countplot(x='Sex', data=df)
```

- More **male** passengers (577) than female (314)

Age Distribution

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```
sns.histplot(df['Age'], bins=30, kde=True)
```

- Ages concentrated between **20–40 years**
-

📌 6. Bivariate Analysis

🎯 Survival by Sex

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```
sns.barplot(x='Sex', y='Survived', data=df)
```

- **Female** survival rate $\approx 74\%$
- **Male** survival rate $\approx 19\%$

🎯 Survival by Pclass

python

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```
sns.barplot(x='Pclass', y='Survived', data=df)
```

- **1st Class:** $\sim 63\%$ survived
- **3rd Class:** $\sim 24\%$ survived

🎯 Survival by Age

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```
sns.kdeplot(df[df['Survived'] == 1]['Age'], label='Survived')
```

```
sns.kdeplot(df[df['Survived'] == 0]['Age'], label='Not Survived')
```

- Higher survival among **children (<10)**
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📌 7. Feature Engineering

👨👩👧👦 Family Size

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```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

- Larger families had **lower survival rates**

Is Alone

python

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```
df['IsAlone'] = (df['FamilySize'] == 0).astype(int)
```

- **Alone passengers** had a **lower survival rate**

Age Groups

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```
df['AgeGroup'] = pd.cut(df['Age'], bins=[0,12,18,35,60,100],  
                        labels=['Child','Teen','YoungAdult','Adult','Senior'])
```

- **Children** had the **highest survival**

8. Correlation Heatmap

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```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
```

- Fare and Pclass are correlated
- Survived correlates positively with Fare and negatively with Pclass

9. Insights

- **Women and children** were prioritized for rescue.
- **1st class passengers** had significantly higher survival chances.
- **Passengers with no family** had lower survival.
- **Younger passengers** were more likely to survive.